Journal of Cancer Research and Clinical Oncology
https://doi.org/10.1007/s00432-018-2718-1

REVIEW – CLINICAL ONCOLOGY

Lentinan as an immunotherapeutic for treating lung cancer: a review of 12 years clinical studies in China

Yiran Zhang[1] · Meng Zhang[1] · Yifei Jiang[1,2] · Xiulian Li[1,2] · Yanli He[1,2] · Pengjiao Zeng[1] · Zhihua Guo[2] · Yajing Chang[1,2] · Heng Luo[1,2] · Yong Liu[1,2] · Cui Hao[1] · Hua Wang[1] · Guoqing Zhang[1] · Lijuan Zhang[1]

Received: 28 May 2018 / Accepted: 19 July 2018
© Springer-Verlag GmbH Germany, part of Springer Nature 2018

Abstract

Purpose Lentinan is a polysaccharide extracted from *Shiitake* mushrooms that have been used to improve general health for thousands of years in Asia. Lentinan injection is a clinically approved drug in several countries in Asia. The purpose of this study is to review the structure, preclinical and clinical studies, and molecular mechanisms of lentinan. Most importantly, the clinical effectiveness of lentinan as an adjuvant therapeutic drug in treating patients with lung cancer in China during the past 12 years is analyzed statistically.

Methods We carried out literature search of randomized controlled trials (RCTs) published from 2004 to 2016 based on CNKI (China National Knowledge Infrastructure), VIP (Chongqing VIP Chinese Scientific Journals Database) and Wanfang database, and 38 eligible RCTs of lentinan-associated lung cancer treatment were identified, containing 3,117 patients.

Results The structure and function relationship and underlying molecular mechanism of lentinan as an immunostimulant has been summarized. The mean value of overall response rate in treating lung cancer was increased from 43.3% of chemotherapy alone to 56.9% of lentinan plus chemotherapy [$p < 0.001$, 95% confidence interval (CI) 0.102–0.170]. Compared with chemotherapy alone, lentinan plus chemotherapy showed more efficacy in treating lung cancer (pooled RR 0.79, 95% CI 0.74–0.85) and no statistical heterogeneity was found among studies ($I^2 = 11\%$).

Conclusion Clinical data presented in the past 12 years shows that lentinan is effective not only in improving quality of life, but also in promoting the efficacy of chemotherapy during lung cancer treatment.

Keywords Cancer · Chemotherapy · Immunobalance · Immunostimulant · Lentinan

Background

In 2012, according to the World Health Organization (WHO), there are about 8.2 million cancer deaths, or 14.6% of deaths worldwide; lung cancer being the deadliest form (Forman and Ferlay 2014; The Top 10 Causes of Death 2018). Progress has been made in cancer treatment (Banin Hirata et al. 2014), but there are still no effective drugs for treating most types of solid cancers (Li 2001). Despite low effectiveness, chemotherapy and radiation therapy are still major treatments for cancer patients (Thomas et al. 2014). Patients have to face severe side effects, and for breast and non-small cell lung cancer, the 5-year survival rate is less than 50% (Barrett 2010; Makanjuola et al. 2014; Schiller et al. 2002). Thus, reducing side effects of cancer treatment and enhancing the quality of life for cancer patients are crucial.

The medicinal qualities of *Shiitake* mushrooms have been known for thousands of years in China (Chen and Seviour 2007; Chen and Raymond 2008). The anti-tumor property of lentinan was reported by Chihara et al. (1969). Lentinan is mainly composed of β-glucan with anti-tumor, anti-inflammatory (Ruthes et al. 2016), anti-diabetes properties (Wang et al. 2016), and other therapeutic properties. Lentinan was approved as an adjuvant for stomach cancer therapy in Japan in 1985. It is

Yiran Zhang and Meng Zhang contributed equally to this review.

✉ Lijuan Zhang
zhanglj@qduhospital.cn

[1] Systems Biology and Medicine Center for Complex Diseases, Affiliated Hospital of Qingdao University, Qingdao 266000, China

[2] School of Medicine and Pharmacy, Ocean University of China, Qingdao 266000, China

approved for treating multiple types of cancer, hepatitis, and other diseases in China. Lentinan is available as capsules, tablets, and injections. The capsules and tablets of lentinan is taken orally as a traditional medicine. Intravenous injection of lentinan is clinically approved and used in hospital with a dose of 1–1.5 mg/day/patient. Clinical data show that lentinan is a biological response modifier and an immunostimulant with proven efficacy in treating hepatitis (Wang 1994; Wu et al. 1993), HIV (Gordon et al. 1998), malignant pleural effusion (Yoshino et al. 2010; Chang et al. 2013) and cancers (Ina et al. 2013; Wang et al. 2012).

Structure of lentinan

There have been numerous reports on structures and components in lentinan, but only β-glucan has been established to be associated with immunologic competence (Murphy et al. 2010). The primary structure of β-glucan in lentinan was demonstrated in 1977 (Saitô et al. 1977), which consists of a β-(1–3)-glucose backbone with two (1–6)-β-glucose branches of every five glucose units (Lv et al. 2009) (Fig. 1). The secondary structure of β-glucan is a single helical conformation confirmed by ^{13}C-NMR (Lv et al. 2009). Surenjav et al. (2006) studied four types of polysaccharides extracted from *Lentinus edodes* (fruiting body). All of them are β-glucans containing 4.6–15.2% proteins and displaying anti-tumor activities. However, the activity is reduced after the polysaccharides are processed by ultrasound and deproteinization. This indicates that the secondary structure, relative molecular mass, and associated proteins may also affect the anti-tumor activities of lentinan.

Immunostimulatory mechanisms of lentinan

Studies (Brown and Gordon 2003) show that β-glucan as the main component of lentinan plays a key role in immunoregulation of lentinan. It can trigger downstream signaling, such as MAPK-NFκB and Syk-PKC, through binding with pattern recognition receptors (TLR2/4/6/9, Dectin-1, etc.), the complement receptor (CD11b), and other membrane receptors (Zhou et al. 2014), which activates the function of immunocytes (NK, macrophage, T cells) (Dai and Wu 1998; Liu et al. 2003). Peng et al. (2008) used lentinan (self-made) to stimulate mouse spleen lymphocytes and the results showed that the expression of both TLR4 and TLR9 are up-regulated initially in addition to up-regulated TNF-α expression by T-lymphocytes. Wang et al. (2008) reported that lentinan (Zhejiang Fangge Pharmaceutical CO., LTD) promotes IgG secretion by B-lymphocytes and enhances phagocytic activity of macrophages in mice. Peter et al. (1988) showed that lentinan (self-made) activates natural killer cells (NK) and induces antibody-dependent cell-mediated cytotoxicity (ADCC) in vitro. Additionally, Kupfahl et al. (2006) reported that pre-treatment of mice with lentinan (self-made) results in increased concentrations of TNF-α, IL-12 and IFN-γ (Fig. 2). Yoshino et al. found that the percentage of CD4+ IFN-γ+ T cell increases significantly ($p < 0.05$) whereas the percentages of CD4+ IL-4+ T cell and CD4+ IL-6+ T cell decreases significantly ($p < 0.02$) in patients' blood with digestive cancers after intravenous administration of lentinan, which indicates that lentinan can reverse the Th2-dominated immune microenvironment of cancer patients, and enhance the ratio of CD8+ T cells and Th1 cells in the whole T cell subgroups (Yoshino et al. 2000; Wang et al. 2010).

Another significant influence of lentinan in regulating immunity may be due to "trained immunity", which was

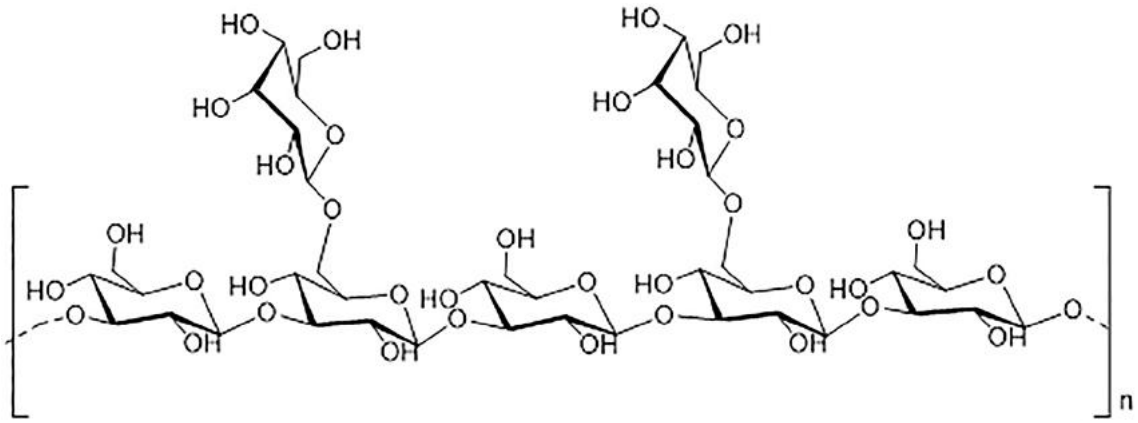

Fig. 1 A schematic diagram of β-glucan structure in lentinan. Glucose units are linked through β-(1–3) glycosidic bond to form backbone with β-(1–6) glucose branches

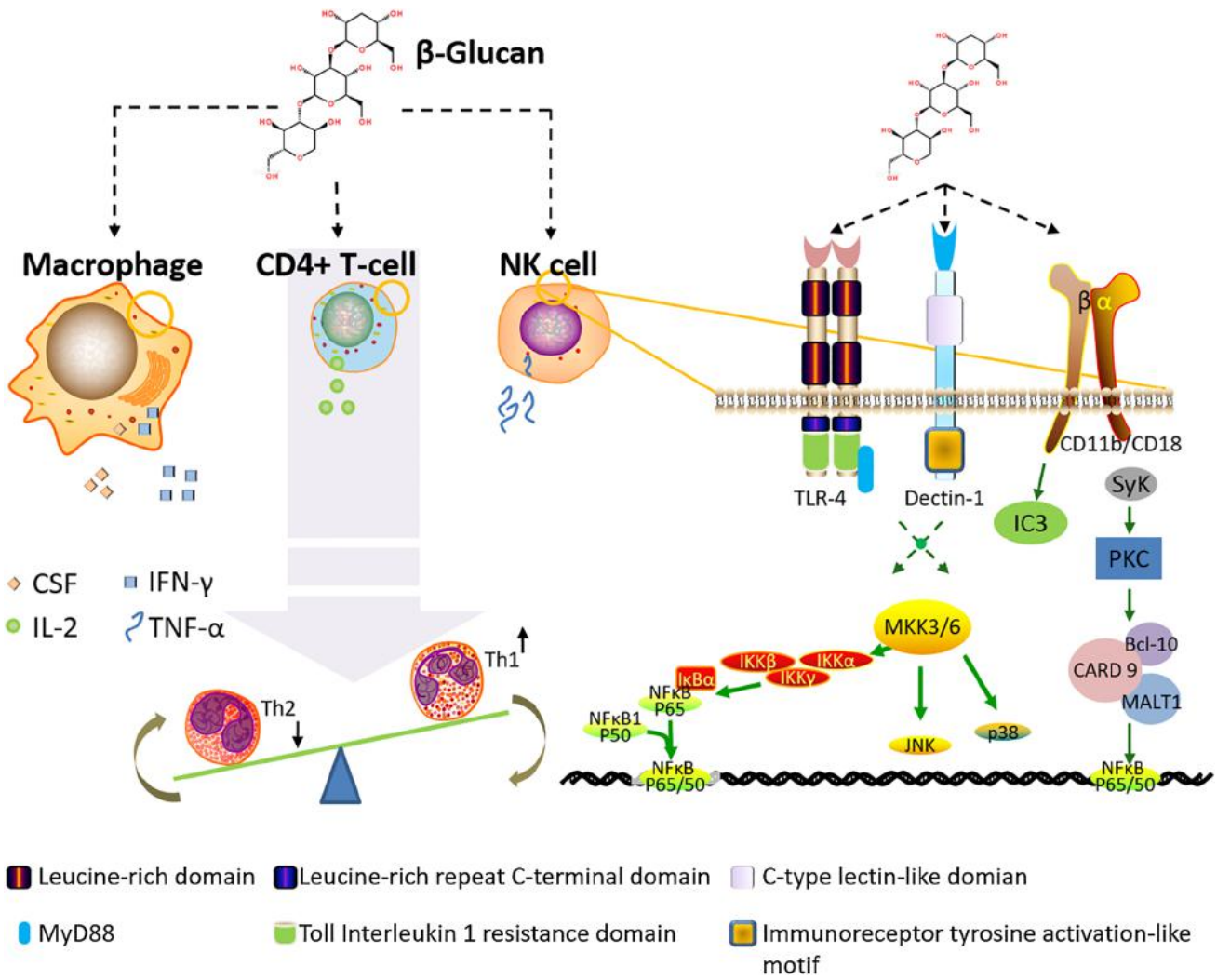

Fig. 2 Immunostimulating and immunobalancing effects of β-glucan. Immunocytes can be activated by β-glucan through binding with membrane receptors, such as pattern recognition receptors (TLR4, Dectin-1), complement receptor (CD11b), etc. Subsequently, downstream signaling pathways, MAPK-NFκB and Syk-PKC-NFκB, are involved in immunoregulation

firstly described as triggered by human pathogen *Candida albicans* (Quintin 2018). As a major component of fungi, β-glucan functions as a stimulus to induce a non-specific memory resulting in an enhanced immune status. Once cells of the innate immune compartments such as monocytes or natural killer (NK) cells encounter a secondary stimulus, an optimal response will be triggered. Studies have shown that trained immunity can improve host defense and result in a better survival (Bekkering et al. 2016; Quintin et al. 2012). The potential molecular mechanisms underlying the induction of innate immune memory by β-glucan is a complex interaction between immunological, metabolic and epigenetic changes. Cheng et al. found that oxygen consumption of monocytes shifted from oxidative phosphorylation to aerobic glycolysis after treating with β-glucan (Cheng et al. 2014). Additionally, β-glucan affects the tri-methylation of the lysine 4 on the histone 3 (H3K4), and reverses the epigenetic state of LPS-induced immunological tolerance (Novakovic et al. 2016).

Cytotoxicity of lentinan

Lentinan can also possess direct cytotoxicity on tumor cells. Bao et al. proved that lentinan could trigger apoptosis through intracellular reactive oxygen species (ROS) while treating human bladder cancer T24 cells (Bao et al. 2015). Similarly, it has been reported that lentinan induced cytotoxicity on S180 cells via mitochondria pathways by up-regulating Bax and down-regulating Bcl-2, which resulted in apoptosis of S180 cells (Zhang et al. 2015a).

Preclinical studies of combination of lentinan with chemotherapy and targeted therapy

Lentinan has been proved to function as a biological response modifier (BRM) during preclinical pharmacological studies (Wang et al. 2015; Nishitani et al. 2013; Xu et al. 2011). It exhibits a great potential to boost immune response to tumor burden. Oncologists realize that such an effective but low toxic immunomodulator is an ideal agent to combine with chemotherapeutics due to the severe side effects and immunosuppressive environment caused by chemotherapy. For treating non-small cell lung cancer (NSCLC), lentinan was found to be synergistic with paclitaxel by activating TXNIP-NLRP3 inflammasome through the ASK1/p38 MAPK signal pathway (Liu et al. 2015). Cisplatin is the first-line drug in treating advanced lung cancer. However, nephrotoxicity is the complication caused by cisplatin. A new study showed that lentinan significantly prevented cis-DDP-induced kidney injury in vivo through activation of the NRF2-ARE signaling pathway (Chen et al. 2016).

Furthermore, targeted therapies that specifically inhibit the growth of cancer cells have been developed greatly (Carter 2001; Clynes et al. 2000). HER2 is a member of the ErbB family that plays an important role in promoting oncogenic transformation and tumor growth (Slamon et al. 1987). Herceptin (trastuzumab) has been successfully used in treating HER2 overexpressed breast cancer patients (Hamy et al. 2018). An in vivo study clearly demonstrated that lentinan significantly suppresses tumor growth combined with Herceptin in BT474 xenografts in nude mice (Cheung et al. 2002). Considering that in approximately 18–33% NSCLC tumors, 17–42% adenocarcinomas, and 2–40% large-cell carcinomas lung cancer demonstrate overexpressed HER2 (Hirsch et al. 2002), the synergistic action with targeting cancer therapy might be expected when lentinan is used in combination with antibodies plus cytotoxic chemotherapeutic agents for patients with HER-2 positive lung cancer.

Clinical effects of lentinan verified by statistical analysis

To explore the effects of chemotherapy plus lentinan (C + L) comparing to that of chemotherapy alone (L), we searched all published clinical studies of lentinan in treating lung cancer during the past 12 years (2004–2016) in China. Our search was conducted using CNKI, VIP and Wanfang databases with the terms of "lung cancer", "lung neoplasms", "lentinan", "chemotherapy" and "randomized controlled trials (RCTs)". Qualified RCTs should report the relative risk (RR) and the corresponding 95% confidence intervals (CIs) or sufficient information to calculate them. As a result, 38 RCTs containing 3117 patients were identified.

Our statistical analyses were conducted by REVMAN Software (version 5.1.2; The Cochrane Collaboration, Oxford, UK) and SPSS (version 19, IBM, Chicago, US). A pooled relative risk (RR) was produced with 95% CIs to summarize the effect of each comparison tested, using a random effects model as a conservative estimate. Global statistical heterogeneity across all comparisons was assessed using the I^2 measure from the 'netmeta' statistical package. The I^2 measure ranges between 0 and 100% and is typically considered low, moderate and high for values of 25–9, 50–74 and ≥ 75%, respectively (Table 1).

Response rates (RR) known as common evaluation indicator of tumor treatment is the sum of complete response (CR) and partial response (PR). Response rate for chemo (C) or chemo + lentinan (C + L) treated patients are summarized and shown in Table 2 with the p values calculated and indicated for each RCT. The reported overall response rates in C + L groups improved significantly compared to C groups (mean value 56.9 vs 43.3%, $p < 0.001$, 95% CI 0.102–0.170). C + L therapy showed more efficacy in treating lung cancer (pooled RR 0.79, 95% CI 0.74–0.85) and no statistical heterogeneity was found among studies ($I^2 = 11\%$) (Fig. 3). Above analyses indicate that lentinan is effective not only on improving quality of life but also on promoting the efficacy of chemotherapy during lung cancer treatment.

The side effects of lentinan

Although the side effects of lentinan is rare, some clinical cases reported complications (Table 3). Low blood pressure, allergy, urgent appeal, and dizziness were the major side effects. Usually, these symptoms would resolve within 15–30 min after drug withdrawal, oxygen uptake, intravenously given 10 mg dexamethasone, or intravenously given 10 ml 10% calcium gluconate. These cases indicate that more observation during nursing is required for the drug, as they have not been tested for anaphylaxis in advance, and only that can avoid serious consequences in time.

Discussion

This article briefly introduced the structure and molecular mechanisms of lentinan in treating lung cancer. Edible mushrooms have been used as traditional herbal medicine in China for thousands of years and it is known to enhance

Table 1 List of abbreviations

Abbr.	Full name	Abbr.	Full name
Drugs			
5-Fu	5-Fluorouracil	IFO	Ifosfamide
ADM	Adriamycin	L-OHP; OX	Oxaliplatin
CAPE	Capecitabine/xeloda	MA	Megestrol acetate
CBP	Carboplatin	MMC	Mitomycin
CF	Leucovorin; calcium folinate	NDP	Nedaplatin
Chemo	Chemotherapy	NVB	Vinorelbine
CS	Cefalothin sodium	PDN	Prednisone
CTX	Adenosine cyclophosphate	PTX	Paclitaxel; taxinol
DDP	Cisplatin; diamminedichloroplatinum	S-1	Tegafur gimeracil oteracil
DXL	Docetaxel; taxotere	THP	Pirarubicin
EPI	Epirubicin	VCR	Vincristine
FT	Tegafur/futrafur	VP-16	Etoposide
GEM	Gemcitabine	YS	Yanshu injection
Drug combinations for chemo			
CAF	CTX + ADM + 5-Fu	GP	GEM + DDP
CAP	CTX + ADM + DDP	IEP	IFO + VP-16 + DDP
CF	CF + 5-Fu	LFP	CF + 5-Fu + DDP
CFOXFT	CF + L-OHP + FT	MAFP	MMC + ADM + 5-Fu + CBP
CHOP	CTX + ADM/THP + VCR + PDN	MATP	MA + PTX + DDP
DCF	DXL + DDP + 5-Fu	MFP	MMC + 5-Fu + DDP
DF	DXL + 5-Fu	NOX	NVB + L-OHP
DOX	DXL + L-OHP	NP	NVB + DDP
DP	DXL + DDP	OF	L-OHP + 5-Fu
ECF	EPI + DDP + 5-Fu	PC	CBP + PTX
ELF	VP-16 + CS + 5-Fu	PCF	PTX + DDP + 5-FU
ELFP	VP-16 + CS + 5-Fu + DDP	PF	PTX + 5-Fu
EOF	EPI + L-OHP + 5-Fu	PN	PTX + NDP
EOX	EPI + L-OHP + CAPE	SOX	S-1 + L-OHP
EP	VP-16 + DDP	SP	S-1 + DDP
FAC	5-FU + ADM + CTX	TA	PTX + ADM
FOLFOX	CF + 5-Fu + L-OHP	TACE	EPI + MMC + CBP
FOLPCF	CF + PTX + DDP + 5-Fu	TP	PTX + DDP
FP	5-Fu + DDP	XELOX	L-OHP + CAPE
FTP	FT + DDP	XrayDP	X-Ray + DXL + DDP

body's immunity. Pharmacological studies show that lentinan is the major biological component in mushrooms. Clinical data presented in the past 12 years show the effect of lentinan on improving quality of life and on promoting the efficacy of chemotherapy and radiation therapy during lung cancer treatment.

Even though it shows cytotoxicity on cancer cells in vitro, lentinan functions mainly as an immune system modulator. The immune system modulator is the earliest method used in cancer treatment that can be dated back to 1892 (Levine 2008). Compared to current immunotherapies, such as immune checkpoint inhibitors (like PD1/PD-L1, CTLA-4), adoptive T-cell therapies (like CAR-T, TCR-T), antibody-dependent treatments (like trastuzumab, rituximab) and cancer vaccines (like HPV-vaccine, Provenge), the anti-tumor effect of lentinan is non-specific and wide spectrum. It is always used as an adjuvant therapy in treating cancer. In the future, the combination of non-specific and specific immunotherapies or the use of different immune system regulators is worth exploring.

Presently, the ability of immunoregulation has drawn more attraction to oncologists and pharmacologists. Imbalance of Th1/Th2 and immunosuppressive tumor microenvironment caused by chemo have been found to be an important reason for failure of chemotherapy (van Meir et al. 2017; Zheng et al. 2017). Accordingly, chemoimmunotherapy has been used to address the shortcomings of chemotherapy (Kay et al. 2017; Ina and Furuta 2017). Glycan-based drugs,

Table 2 Response rate for chemo (C) or chemo + lentinan (C + L) treated patients with lung cancer

Cases (M/F)	Age	C group cases (M/F)	drugs used	C + L group cases (M/F)	drugs used	Dosage and administration of lentinan	Duration (weeks)	(C + L)/C (%) response rate	References
112 (75/37)	35–71	56 (38/18)	XrayDP	56 (37/19)	L + XrayDP	1 mg/d, i.v	4	88.7/83.3	Bai (2010)
60 (39/21)	55 ± 3	30 (20/10)	GP	30 (19/11)	L + GP	1 mg/d, i.v	4	86.7/76.7	Yin and Yang (2016)
46 (29/17)	59[a]	20 (12/8)	DDP	26 (17/9)	L + DDP	1 mg/d, i.v	6	84.6/55.0*	Lu (2013)
140 (78/62)	60–85	70 (38/32)	GP/TP	70 (40/30)	L + GP/TP	1.5 mg/d, i.v	8	82.9/62.9*	Zhang (2016)
49	N/A	26	EP	23	L + EP	1 mg/d, i.v	6	76.9/39.1**	He et al. (2015)
104 (54/50)	33–73	52 (27/25)	NP	52 (27/25)	L + NP	1.5 mg/d, i.v	6	71.2/44.2*	Ma (2014)
86 (44/42)	N/A	42 (25/17)	NP	44 (28/16)	L + NP	1.5 mg/d, i.v	8	70.5/69.0	Ding et al. (2012)
98 (65/33)	60 ± 13	49 (34/15)	GP	49 (31/18)	L + GP	1.5 mg/d, i.v	4	69.4/55.1*	Li et al. (2016)
139 (80/59)	60–86	70 (38/32)	TP/GP	69 (42/27)	L + TP/GP	1.5 mg/d, i.v	6	66.7/60.0	Zhang et al. (2015b)
87 (57/30)	66 ± 14	46 (32/14)	EP	41 (25/16)	L + EP	1.5 mg/d, i.v	6	65.9/43.5**	Song and Zhu (2016)
68 (42/26)	25–78	34	GP	34	L + GP	1.2 mg/d, i.v	8	64.7/47.1*	Dai (2014)
62 (38/24)	38–69	31 (18/13)	GP	31 (20/11)	L + GP	1 mg/d, i.v	8	64.5/38.7*	Li et al. (2009)
61 (39/22)	39–74	30 (20/10)	GP	31 (19/12)	L + GP	1 mg/d, i.v	5	64.5/36.7*	Zhao and Ma (2011)
66 (37/29)	N/A	32 (18/14)	DP	34 (19/15)	L + DP	1 mg/d, i.v	4	61.7/31.2*	Qian (2010)
65 (51/14)	43–71	32 (26/6)	CAP + CE	33 (25/8)	L + CAP + CE	1 mg/d, i.v	4	57.6/34.4*	Chai and Xiao (2005)
231 (146/85)	60 ± 14	105 (68/37)	GP	126 (78/48)	L + GP	1 mg/d, i.v	6	57.1/40.0*	Wu (2015)
52 (31/21)	45–78	22 (13/9)	GP	30 (18/12)	L + GP	1.5 mg/d, i.v	6	33.3/54.5*	Li et al. (2013)
52 (37/15)	55–75	26 (20/6)	PC	26 (17/9)	L + PC	1 mg/d, i.v	8	53.8/42.3	Dai (2010)
63 (38/25)	43–71	31	NP	32	L + NP	1.5 mg/d, i.v	8	40.6/48.4	Lu et al. (2008)
68	32–74	34	GP/NP/TP	34	L + GP/NP/TP	1 mg/d, i.v	4	52.9/44.1	Zhao (2012)
100 (72/28)	25–80	50	GP	50	L + GP	1 mg/d, i.v	5	52.0/28.0**	Zhu and Zhang (2013)
90 (62/28)	≤ 75	45 (30/15)	GP/NP/DP	45 (32/13)	L + GP/NP/DP	1 mg/d, i.v	6	51.1/28.9**	Zhao et al. (2011)
76 (51/25)	40–60	38 (25/13)	MATP	38 (26/12)	L + MATP	1.5 mg/d, i.v	6	50.0/47.4	Xie et al. (2006)
78 (66/12)	34–71	26 (22/4)	NOX	52 (44/8)	L + NOX	1.5 mg/d, i.v	4	50.0/46.1	Shang and Huang (2009)
81 (65/16)	18–60	39 (32/7)	NP	42 (33/9)	L + NP	1 mg/d, i.v	2	50.0/33.3**	Wang et al. (2006)
98 (64/34)	26–78	49	GP	49	L + GP	1 mg/d, i.v	4	49.0/40.8	Jiang et al. (2010)
70	32–72	35	GP/NP/DP	35	L + GP/NP/DP	1.5 mg/d, i.v	4	48.6/40.0	Jiang and Zhou (2009)
64 (55/9)	35–69	31 (27/4)	NP	33 (28/5)	L + NP	1.2 mg/d, i.v	6	48.5/35.5*	Shi et al. (2007)
61 (45/16)	65–85	29 (22/7)	NP	32 (23/9)	L + NP	1 mg/d, i.v	8	46.9/34.5*	Wu (2009)
69	N/A	34	GP	35	L + GP	1 mg/d, i.v	6	45.7/41.2	Zhang et al. (2014)
68 (39/29)	35–76	35	PC	33	L + PC	1 mg/d, i.v	4	45.5/45.7	Yue and Jia (2010)
113 (68/45)	42–76	56	DP	57	L + DP	1.5 mg/d, i.v	5	43.9/37.5	Zhang et al. (2009)
80 (58/22)	33–74	40 (30/10)	NP	40 (28/12)	L + NP	1 mg/d, i.v	6	42.5/45.0	Xin et al. (2016)
82 (63/19)	51 ± 4	42 (33/9)	GP	40 (30/10)	L + GP	1.5 mg/d, i.v	8	42.5/41.5	Wang et al. (2011)
86 (55/31)	43–72	43 (26/17)	DP	43 (29/14)	L + DP	1.5 mg/d, i.v	8	37.2/16.3*	Wei and Xie (2015)
71(50/21)	26–74	36 (26/10)	GP	35 (24/11)	L + GP	1 mg/d, i.v	6	37.1/36.1	Zhao et al. (2013)
51 (29/22)	56–58[a]	26 (14/12)	GP	25 (15/10)	L + GP	1 mg/d, i.v	4	36.0/34.6	Han et al. (2012)
80 (53/27)	18–70	40 (28/12)	DP	40 (25/15)	L + DP	1 mg/d, i.v	6	30.0/15.0*	Xiao et al. (2015)

N/A not available

**$p < 0.01$, *$p < 0.05$, Student's *t* test

[a]Median

such as lentinan, are potential and ideal immunoregulating drugs that can be combined with chemotherapeutics due to their broad spectrum of therapeutic properties, relatively low toxicity, and low costs. In fact, other types of tumors have been treated with lentinan and chemotherapeutics. For example, a phase III study comparing therapy using S-1/lentinan with S-1 alone is now underway in Japan (Ina et al. 2011). We deem that these clinical experiences may be helpful in treating lung cancer in future and increasing full use of lentinan for cancer patients worldwide.

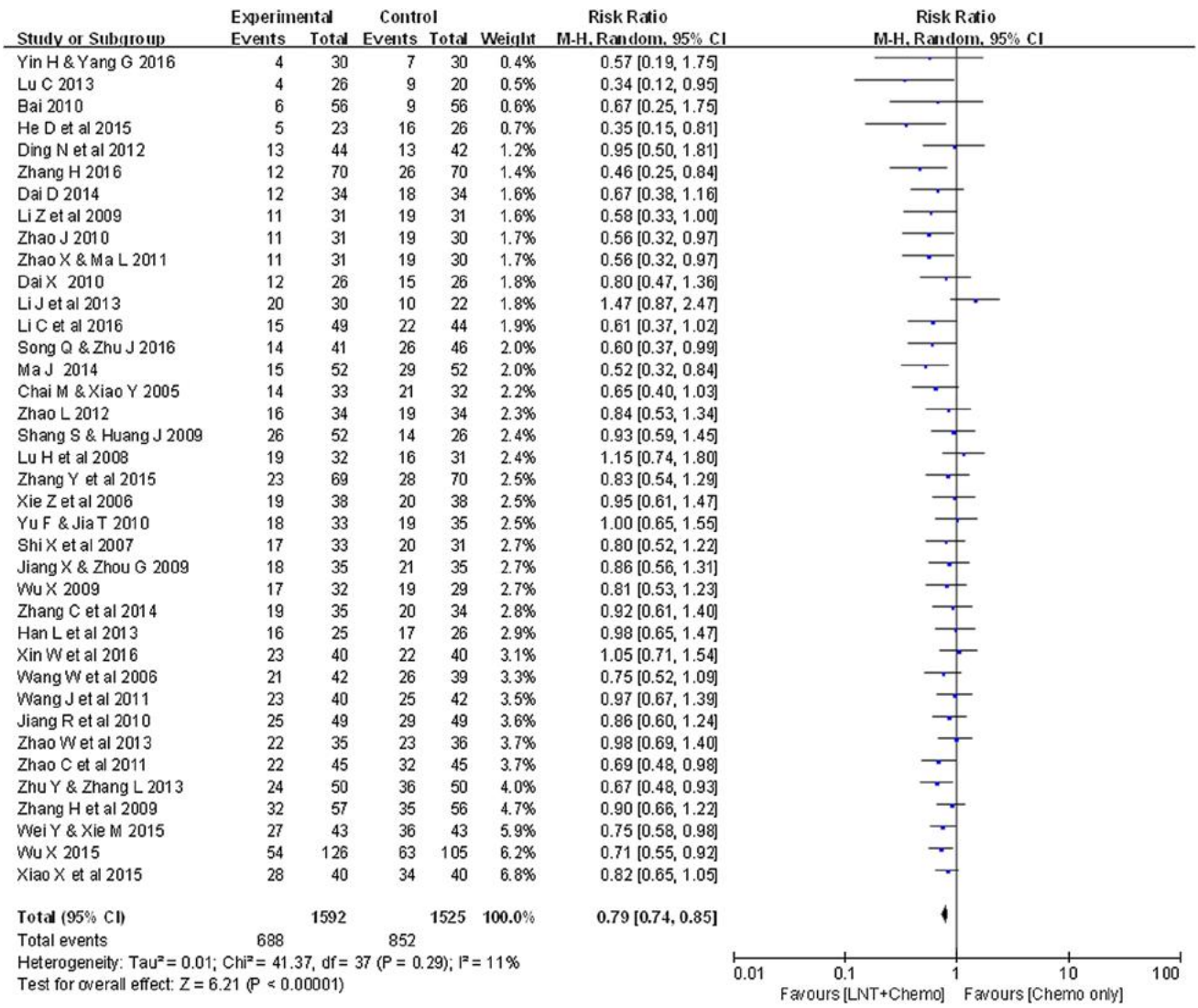

Fig. 3 Forest plot of summary relative risks (RRs) of lentinan-associated chemotherapy and chemotherapies. Lentinan adjuvant therapy was ranked more effective in treating lung cancer (RR 0.79, 95% CI 0.74–0.85). I^2 for global statistical heterogeneity = 11%

Table 3 Lentinan-associated adverse reactions in treating lung cancer

Cases (M/F)	Age	Adverse events	References
14 (8/6)	36–82	Low blood pressure, allergic shock	Zhou and Liu (2010)
1 (1/0)	51	Chest tightness, palpitation, dyspnea, blurred mind, hypotension	Deng et al. (2010)
3 (1/2)	46–56	Dizziness, low blood pressure, chilly	Peng et al. (2013)

Acknowledgements The authors thank Alexander Lu of St. Louis University for language editing of the entire manuscript.

Funding This research was supported by National Natural Science Foundation of China (Grant nos. 91129706, 81672585); Key Technology Fund of Shandong Province (Grant 2016ZDJS07A07); the "Double First-Class" fund of Shandong Province; Taishan Scholar Fellowship (05052011) and Postdoctoral application research project in Qingdao, China (Grant 40518060042).

Compliance with ethical standards

Conflict of interest Authors declare no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

References

Bai R (2010) Radiation and chemotherapy combined with lentinan to treat with non-small cell lung cancer with brain metastasis. Hebei Med J 32:2046–2047

Banin Hirata BK, Oda JMM, Losi Guembarovski R et al (2014) Molecular markers for breast cancer: prediction on tumor behavior. Dis Mark 2014:1–12

Bao L, Wang Y, Ma R et al (2015) Apoptosis-inducing effects of lentinan on the proliferation of human bladder cancer T24 cells. Pak J Pharm Sci 28:1595–1600

Barrett SV (2010) Breast cancer. J R Coll Physicians Edinb 40:335–339

Bekkering S, Blok BA, Joosten LA et al (2016) In vitro experimental model of trained innate immunity in human primary monocytes. Clin Vaccine Immunol 23:926–933

Brown GD, Gordon S (2003) Fungal beta-glucans and mammalian immunity. Immunity 19:311–315

Carter P (2001) Improving the efficacy of antibody-based cancer therapies. Nat Rev Cancer 1:118–129

Chai M, Xiao Y (2005) Effect of lentinan in the treatment of advanced non-small cell lung cancer. J Med Forum 26:69–70

Chang X, Shao P, Huang X et al (2013) Clinical research on cisplatin combined lentinan in local therapy of malignant pleural effusion. New Med 44:840–842

Chen J, Raymond K (2008) Beta-glucans in the treatment of diabetes and associated cardiovascular risks. Vasc Health Risk Manag 4:1265–1272

Chen J, Seviour R (2007) Medicinal importance of fungal β-(1 → 3), (1 → 6)-glucans. Mycol Res 111:635–652

Chen Q, Peng H, Dong L et al (2016) Activation of the NRF2-ARE signalling pathway by the Lentinula edodes polysaccharose LNT alleviates ROS-mediated cisplatin nephrotoxicity. Int immunopharmacol 36:1–8

Cheng SC, Quintin J, Cramer RA et al (2014) mTOR- and HIF-1α-mediated aerobic glycolysis as metabolic basis for trained immunity. Science 345(6204):1250684

Cheung NK, Modak S, Vickers A et al (2002) Orally administered beta-glucans enhance anti-tumor effects of monoclonal antibodies. Cancer Immunol Immunother CII 51:557–564

Chihara G, Maeda Y, Hamuro J et al (1969) Inhibition of mouse sarcoma 180 by polysaccharides from *Lentinus edodes* (Berk.) sing. Nature 222:687–688

Clynes RA, Towers TL, Presta LG et al (2000) Inhibitory Fc receptors modulate in vivo cytotoxicity against tumor targets. Nat Med 6:443–446

Dai X (2010) Observation on chemotherapy combined with lentinan to treat metaphase or terminal non-small cell lung cancer. J Clin Med Pract 14:73, 77

Dai D (2014) Recent and long-term effect and adverse reaction of lentinan combined with GP chemotherapy in the treatment of non-small cell lung cancer. Mod J Integr Tradit Chin West Med 23:1627–1628

Dai J, Wu W (1998) Regulatory effect of lentinan on immune functions of patients with lung cancer. Cancer Res Prev Treat 25:144–145

Deng M, Wang S, Liu Z (2010) One case of anaphylactic shock induced by lentinan injection. Chin J Drug Appl Monit 07:321–322

Ding N, Huang Z, Zhang Q et al (2012) Observation of clinical effect of lentinan and vinorelbine and cisplatin recombination on patients with advanced non-small cell lung cancer. Anhui Med J 33:681–683

Forman D, Ferlay J (2014) The global and regional burden of cancer. In: Stewart BW, Wild CP (eds) World cancer report 2014. International Agency for Research on Cancer, Lyon, p 16

Gordon M, Bihari B, Goosby E et al (1998) A placebo-controlled trial of the immune modulator, lentinan, in HIV-positive patients: a phase I/II trial. J Med 29:305–330

Hamy AS, Belin L, Bonsang-Kitzis H et al (2018) Pathological complete response and prognosis after neoadjuvant chemotherapy for HER2-positive breast cancers before and after trastuzumab era: results from a real-life cohort. Br J Cancer 118:e14

Han L, Li J, Liu Y (2012) A clinical observation of lentinan combined with GP regimen in patients with advanced non-small cell lung cancer. Her Med 31:1029–1032

He D, Li J, Chen Z (2015) Recent and long-term effect observation of lentinan combined with EP chemotherapy in the treatment of small cell lung cancer. J Huaihai Med 33:598–599

Hirsch FR, Franklin WA, Veve R et al (2002) HER2/neu expression in malignant lung tumors. Semin Oncol 29:51–58

Ina K, Furuta R (2017) Complete response of metastatic gastric cancer to chemoimmunotherapy. Indian J Med Res 146:141

Ina K, Furuta R, Kataoka T et al (2011) Lentinan prolonged survival in patients with gastric cancer receiving S-1-based chemotherapy. World J Clin Oncol 2:339–343

Ina K, Kataoka T, Ando T (2013) The use of lentinan for treating gastric cancer. Anticancer Agents Med Chem 13:681–688

Jiang X, Zhou G (2009) Clinical research of combined lentinan for injection with chemotherapy on advanced non-small cell lung cancer. China Mod Dr 47:74–75

Jiang R, Tan X, Liu W et al (2010) Therapeutic effect of combined lentinan, gemcitabine and cisplatin in the treatment of advanced non-small cell lung cancer. Med J Qilu 25:132–134

Kay NE, LaPlant BR, Pettinger AM et al (2018) Cumulative experience and long term follow-up of pentostatin-based chemoimmunotherapy trials for patients with chronic lymphocytic leukemia. Expert Rev Hematol 11:337–349

Kupfahl C, Geginat G, Hof H (2006) Lentinan has a stimulatory effect on innate and adaptive immunity against murine *Listeria monocytogenes* infection. Int Immunopharmacol 6:686–696

Levine DB (2008) The Hospital for the Ruptured and Crippled: William Bradley Coley, third Surgeon-in-Chief 1925–1933. HSS J 4:1–9

Li X (2001) Progress in anti-tumor drugs. China Pharm 12:429–431

Li Z, Zhu Y, Zhang P et al (2009) Clinical observation on GP chemotherapy combined with lentinan to treat advanced non-small cell lung cancer. J Pract Med 25:3480–3482

Li J, Song W, Wang S (2013) 30 cases of advanced non-small cell lung cancer treat with lentinan combined with GP chemotherapy. Guangdong Med J 34:1432–1434

Li C, Zhu T, Lu X (2016) Clinical analysis of the application of lentinus edodes polysaccharides combined with chemotherapy in the treatment of patients with advanced lung cancer. Mod Diagn Treat 27:1614–1616

Liu D, Qian J, Pu M (2003) Research status of anti-tumor effect of lentinan. Edible Fungi 2:6–7

Liu W, Gu J, Qi J et al (2015) Lentinan exerts synergistic apoptotic effects with paclitaxel in A549 cells via activating ROS-TXNIP-NLRP3 inflammasome. J Cell Mol Med 19:1949–1955

Lu C (2013) Clinical observation on 26 cases of lung cancer with malignant pericardial effusion treated with lentinan and cisplatin. Guiding J Tradit Chin Med Pharm 19:53–54

Lu H, Hong S, Zhang X (2008) Clinical analysis of 63 cases with advanced non-small cell lung cancer treated with lentinan combined with NP chemotherapy. J XinJiang Med Univ 31:92–93

Lv G, Fan L, Zhang Z et al. (2009) Development of research on lentinan. Acta Agriculturae Zhejiangensis 21:183–188

Ma J (2014) Efficacy of lentinan combined with NP regimen in treatment of patients with advanced non-small cell lung cancer and its effect on immunologic function. Chin J Biochem Pharm 34:154–156

Makanjuola SB, Popoola AO, Oludara MA (2014) Radiation therapy: a major factor in the 5-year survival analysis of women with breast cancer in Lagos. Niger Radiother Oncol 111:321–326

Murphy EA, Davis JM, Carmichael MD (2010) Immune modulating effects of beta-glucan. Curr Opin Clin Nutr Metab Care 13:656–661

Nishitani Y, Zhang L, Yoshida M et al (2013) Intestinal anti-inflammatory activity of lentinan: influence on IL-8 and TNFR1 expression in intestinal epithelial cells. PLoS One 8:e62441

Novakovic B, Habibi E, Wang SY et al. (2016) β-Glucan reverses the epigenetic state of LPS-induced immunological tolerance. Cell 167(5):1354–1368

Peng X, Fan X, Shi X et al (2008) Influence of lentinan on expression of TLR4, TLR9 of mouse splenic T/B cells. Chin J Clin Pharmacol 24:234–236

Peng J, Zhang X, Chen C et al (2013) Nursing care of 3 cases of allergy caused by intravenous infusion of lentinan. Chin Gen Pract Nurs 11:1512

Peter G, Karoly V, Imre B et al (1988) Effects of lentinan on cytotoxic functions of human lymphocytes. Immunopharmacol Immunotoxicol 10:157–163

Qian J (2010) 34 cases clinical observation of lentinan adjuvant therapy in the treatment of non-small cell lung cancer. Chin J Mod Drug Appl 04:114–115

Quintin J (2018) Fungal mediated innate immune memory, what have we learned? Semin Cell Dev Biol. https://doi.org/10.1016/j.semcdb.2018.05.023

Quintin J, Saeed S, Martens JHA et al (2012) Candida albicans infection affords protection against reinfection via functional reprogramming of monocytes. Cell Host Microbe 12:223–232

Ruthes AC, Smiderle FR, Iacomini M (2016) Mushroom heteropolysaccharides: a review on their sources, structure and biological effects. Carbohyd Polym 136:358–375

Saitô H, Ohki T, Takasuka N et al (1977) A 13C-N.M.R.-Spectral study of a gel-forming, branched (1 → 3)-β-d-glucan, (lentinan) from *Lentinus edodes*, and its acid-degraded fractions. Structure, and dependence of conformation on the molecular weight. Carbohyd Res 58:293–305

Schiller JH, Harrington D, Belani CP et al (2002) Comparison of four chemotherapy regimens for advanced non-small-cell lung cancer. New Engl J Med 346:92–98

Shang S, Huang J (2009) Observation on short-term therapeutic and adverse effect of lentinan and vinorelbine and oxaliplatin in recombination on patients with advanced non-small cell lung cancer. Clin Focus 24:501–504

Shi X, Ding Q, Yang Q (2007) Lentinan combined with chemotherapy in the treatment of advanced non-small cell lung cancer. China Cancer 16:946–948

Slamon DJ, Clark GM, Wong SG et al (1987) Human breast cancer: correlation of relapse and survival with amplification of the HER-2/neu oncogene. Science (New York, NY) 235:177–182

Song Q, Zhu J (2016) Effect of Mushroom polysaccharide injection on serum tumor markers and clinical effect in patients with stage III and IV SCLC. J China Prescr Drug 14:12–13

Surenjav U, Zhang L, Xu X et al (2006) Effects of molecular structure on antitumor activities of (1 → 3)-β-D-glucans from different *Lentinus edodes*. Carbohyd Polym 63:97–104

The Top 10 Causes of Death (2018). http://www.who.int/mediacentre/factsheets/fs310/en/. Accessed 23 July 2018

Thomas AA, Brennan CW, DeAngelis LM et al (2014) Emerging therapies for glioblastoma. JAMA Neurol 71:1437–1444

van Meir H, Nout RA, Welters MJ et al (2017) Impact of (chemo) radiotherapy on immune cell composition and function in cervical cancer patients. Oncoimmunology 6:e1267095

Wang J (1994) Clinical observation of lentinan in treatment of 60 cases of hepatitis B. J Chin Med 16:23–24

Wang W, Dai X, Ouyang X (2006) Efficacy of lentinan combined with chemotherapy in advanced non-small cell lung cancer. China J Lung Cancer 09:78–81

Wang G, Lin W, Zhao R et al (2008) Effects of six polysaccharides extracted from plants on the immunological cells of mice. J Hyg Res 37:577–580

Wang J, Dong S, Liu C et al (2010) β-Glucan oligosaccharide enhances CD8(+) T cells immune response induced by a DNA vaccine encoding hepatitis B virus core antigen. J Biomed Biotechnol 2010:645213

Wang J, Zhang Y, Zhao H (2011) The influence of immune functions and therapeutic and adverse effect of lentinan combined with chemotherapy in advanced non-small cell lung cancer. Natl Med Front China 06:15–17

Wang JL, Bi Z, Zou JW et al (2012) Combination therapy with lentinan improves outcomes in patients with esophageal carcinoma. Mol Med Rep 5:745–748

Wang Y, Han X, Li YD et al (2015) Effects of tumor-specific antigen induced by lentinan on murine H22 hepatocellular carcinoma immunoprophylaxis. Eur Rev Med Pharmacol Sci 19:4516–4524

Wang P, Zhao S, Yang B et al (2016) Anti-diabetic polysaccharides from natural sources: a review. Carbohyd Polym 148:86–97

Wei Y, Xie M (2015) The influence of lentinan injection on curative effect and quality life of patients with medium and advanced lung cancer after chemotherapy. Guiding J Tradit Chin Med Pharm 21:25–27

Wu X (2009) Clinical observation on advanced non-small cell lung cancer in the old treated with lentinan combined with NVB and DDP. China Foreign Med Treat 28:76–77

Wu X (2015) Clinical analysis of lentinan combined with chemotherapy in treatment of advanced lung cancer. Chin Arch Tradit Chin Med 27:456–458

Wu B, Zhang Y, Zhang L (1993) III clinical trial summary of lentinan in the treatment of 108 cases with chronic viral hepatitis. Fujian J Tradit Chin Med 24:10–13, 50

Xiao X, Liu C, Yuan X et al (2015) Clinical effect of lentinan combined with DP chemotherapy on middle-late stage non-small cell lung cancer. Pract J Card Cereb Pneumal Vasc Dis 23:91–93

Xie Z, Xie J, Wen J et al (2006) A comparison between TP regimen with lentinan and with megestrol acetate for advanced non-small cell lung cancer. China Cancer 15:199–202

Xin W, Shi L, Zhou Y (2016) Efficacy of lentinan combined with NP chemotherapy in treatment of patients with advanced non-small cell lung cancer and its effect on immunologic function. Mod J Integr Tradit Chin West Med 25:3349–3351

Xu X, Pan C, Zhang L et al (2011) Immunomodulatory beta-glucan from *Lentinus edodes* activates mitogen-activated protein kinases and nuclear factor-kappaB in murine RAW 264.7 macrophages. J Biol Chem 286:31194–31198

Yin H, Yang G (2016) Efficacy of synergistic combination of lentinan and GP regimen in stage IV lung squamous cell carcinoma. Oncol Prog 14:687–689

Yoshino S, Tabata T, Hazama S et al (2000) Immunoregulatory effects of the antitumor polysaccharide lentinan on Th1/Th2 balance in patients with digestive cancers. Anticancer Res 20:4707–4711

Yoshino S, Yoshida S, Maeda N et al (2010) Clinical evaluation of the combination treatment of intrapleural or intraperitoneal administration of lentinan and OK-432 for malignant effusion. Gan to kagaku ryoho. Cancer Chemother 37:2798–2800

Yue F, Jia T (2010) Clinical observation of lentinan combined with paclitaxel liposome on advanced non-small cell lung cancer. Pract J Card Cereb Pneumal Vasc Dis 18:1791–1792

Zhang H (2016) Clinical analysis of lentinan combined with chemotherapy for elderly patients with advanced non-small cell lung cancer. Shenzhen J Integr Tradit Chin West Med 26:44–45

Zhang H, Xu J, Ren S (2009) Lentinan combined with chemotherapy in treatment of advanced non-small cell lung cancer. Chin J Cancer Biother 16:523–525

Zhang C, Xie C, Kang Z et al (2014) Clinical observation on GP chemotherapy combined with lentinan to treat metaphase or terminal non-small cell lung cancer. J Qiqihar Univ Med 35:164–165

Zhang Y, Li Q, Shu Y et al (2015a) Induction of apoptosis in S180 tumour bearing mice by polysaccharide from *Lentinus edodes* via mitochondria apoptotic pathway. J Funct Foods 15:151–159

Zhang Y, Yu J, Xiang Y et al (2015b) Clinical efficacy and safety of lentinan combined with chemotherapy for elderly patients with advanced non-small cell lung cancer. Antitumor Pharm 03:222–225

Zhao L (2012) A clinical study on non-small cell lung cancer therapy by combination of lentinan with chemotherapy. Anhui Med Pharm J 16:1496–1497

Zhao X, Ma L (2011) Toxicity-reducing and efficacy-enhancing action of lentinan with chemotherapy in the treatment of advanced non-small cell lung cancer. Clin Ration Drug Use 04:20–21

Zhao C, Li X, Wang Z et al (2011) Immune function and effect of concurrent chemoradiotherapy combined lentinan on patients with advanced non-small cell lung cancer. Anhui Med Pharm J 15:1431–1433

Zhao W, Chen D, Chen J (2013) Curative effect and adverse reaction of lentinan combined with gemcitabine and cisplatin in the treatment of non-small cell lung cancer. Chin J Clin Pharmacol Ther 18:71–77

Zheng H, Zeltsman M, Zauderer MG et al (2017) Chemotherapy-induced immunomodulation in non-small-cell lung cancer: a rationale for combination chemoimmunotherapy. Immunotherapy 9:913–927

Zhou C, Liu M (2010) 30 cases of adverse reaction caused by lentinan. Zhejiang J Integr Tradit Chin West Med 20:635

Zhou Z, Han Z, Zeng Y et al (2014) Chinese FDA approved fungal glycan-based drugs: an overview of structures, mechanisms and clinical related studies. Transl Med 04:1000141

Zhu Y, Zhang L (2013) Short-term curative effect of lentinan combined with chemotherapy in the treatment of lung cancer and its evaluation of immune function. China Pract Med 08:156–157